AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2021

Physics

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

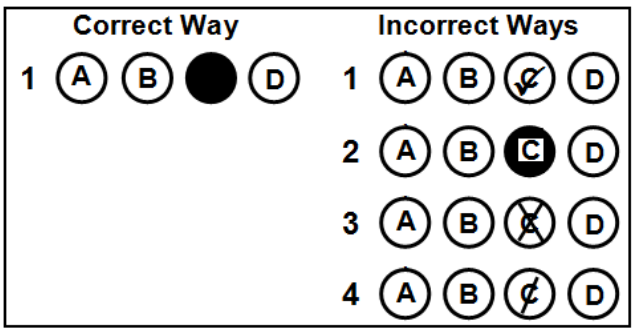
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.

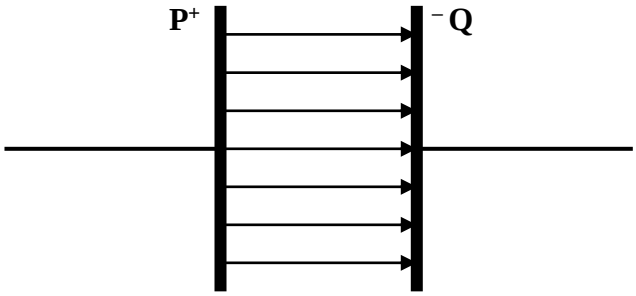


Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. In the given figure, electric field lines of forces between two parallel metallic plates are passing from a positive plate **P** to a negative plate **Q**.



The intensity of the electric field is

A. zero between the plates.
B. stronger near the plate P.
C. stronger near the plate Q.
D. uniform between the plates.

2. If a solid metallic sphere contains positive charges **Q**, then according to Gauss's law, the position of charges **Q** is uniformly

A. concentrated at the centre of the sphere.
B. distributed on the inner surface of the sphere.
C. concentrated inside and outside of the sphere.
D. distributed on the outer surface of the sphere.

3. The total number of electric field lines of forces passing through a unit area is called electric

A. flux.
B. field.
C. intensity.
D. potential.

4. If three capacitors of different capacitances are connected in series, then the quantity that remains the same will be

A. charge.
B. voltage.
C. current.
D. capacitance.

5. Which of the following materials has no free charge?

A. Metals
B. Dielectrics
C. Conductors
D. Semi-conductors

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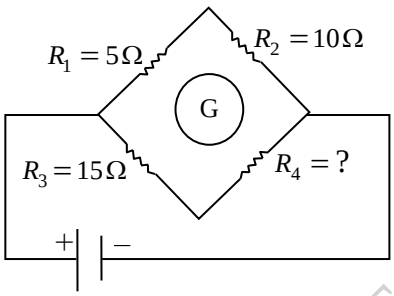
6. Displacement of positive and negative charges in a conductor by an electric field produces

   A. electric flux.
   B. magnetic flux.
   C. electric dipole.
   D. magnetic intensity.

7. If the distance between two identical charges is doubled, then the force of repulsion between them decreases by

   A. two-third of the previous value.
   B. one-third of the previous value.
   C. one-fourth of the previous value.
   D. three-fourth of the previous value.

8. What is the value of R_4 in the given Wheatstone bridge?



   A. 10 Ω
   B. 15 Ω
   C. 30 Ω
   D. 60 Ω

9. A battery of electromotive force 18 V delivers a charge of 1.5 C to an external circuit. The total energy dissipated in the circuit is

   A. 12.0 J.
   B. 16.5 J.
   C. 19.5 J.
   D. 27.0 J.

10. A filament lamp behaves like an Ohmic conductor when its temperature

   A. increases.
   B. decreases.
   C. becomes 0°C.
   D. remains constant.

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11. The given table shows information about two wires, **X** and **Y** that are made of the same material.

Wire	**X**	**Y**
Diameter	2 cm	4 cm
Length	1 m	4 m

Which of the following options is CORRECT for the resistance and resistivity of wires, **X** and **Y**?

	Resistance	**Resistivity**
A	Same	Same
B	Same	Different
C	Different	Same
D	Different	Different

12. The unit for electric current is

A. erg/second.
B. joule/second.
C. dyne/second.
D. coulomb/second.

13. The reciprocal of the resistivity of a material is called

A. permittivity.
B. conductivity.
C. power dissipation.
D. internal resistance.

14. The resistance of a metallic conductor increases with the rise in temperature. This is because the frequency of collisions among the vibrating atoms of the conductor and electrons

A. increases.
B. decreases.
C. becomes zero.
D. remains unchanged.

15. A 1000 W lamp operates at 220 V for 60 s.

The electric current passing through the lamp will be

A. 1.66 A.
B. 3.57 A.
C. 4.54 A.
D. 6.25 A.

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16. The electromotive force of a battery is the voltage between its terminals when

    A. the circuit is closed.
    B. the circuit is opened.
    C. its circuit resistance is minimum.
    D. its internal resistance is maximum.

17. A current carrying coil is placed in a uniform magnetic field.

    The torque produced in this coil will be doubled when the

    A. force in the magnetic field is doubled.
    B. length of coil is reduced to half.
    C. moment arm is reduced to half.
    D. area of coil is doubled.

18. A metallic wire of length 1 m is placed perpendicular to a uniform magnetic field of 10 T.

    If this wire is carrying current of 100 A, then the force acting on this wire will be

    A. 0 N.
    B. 10 N.
    C. 100 N.
    D. 1000 N.

19. An electrical galvanometer becomes more sensitive when

    A. the area of its coil is made small.
    B. it converts into voltmeter and ammeter.
    C. a small amount of current produces larger deflection.
    D. a small amount of current produces smaller deflection.

20. An electron is travelling with a velocity of 10^6 m/s, perpendicular to a field of strength 1 W/m^2. The magnetic force on the electron will be

    (**Note**: Take the charge of an electron as 1.6×10^{-19} C.)

    A. 1.6×10^{-25} N.
    B. 1.6×10^{-13} N.
    C. 1.6×10^{13} N.
    D. 1.6×10^{25} N.

21. When a charged particle moves through a magnetic field, the effect of the field changes the particle's

    A. mass.
    B. volume.
    C. direction of motion.
    D. energy in the system.

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22. Due to an induced electromotive force, current passes through a coil. The amount of this current depends on the

    A. magnetic flux.
    B. area of the coil.
    C. shape of the coil.
    D. resistance of the coil.

23. The process in which a change of current in one coil causes an induced electromotive force in another nearby coil is called

    A. self-induction.
    B. self-inductance.
    C. mutual induction.
    D. mutual inductance.

24. A transformer mainly consists of an iron core and two coils of copper.

    If an additional coil is winded on the iron core of a transformer, then the magnetic flux through it will

    A. increase.
    B. decrease.
    C. become zero.
    D. remain the same.

25. A transformer is used to change the

    I. resistance
    II. capacitance
    III. current
    IV. voltage

    A. I and II.
    B. I and IV.
    C. II and III.
    D. III or IV.

26. The principle of an alternating current generator is based on

    A. Lenz's law.
    B. Faraday's law.
    C. Ampere's law.
    D. Coulomb's law.

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27. Current and voltage in a direct current (DC) circuit are controlled by a/ an

    I. resistor
    II. inductor
    III. capacitor
    IV. transistor

    A. I and II.
    B. I and IV.
    C. II and III.
    D. III and IV.

28. The ratio between the root mean square (rms) of an applied voltage and the rms resulting from an alternating current is called

    A. resistance.
    B. impedance.
    C. capacitance.
    D. conductance.

29. The phase difference between each pair of coils of a three phase alternating current generator is

    A. 0°
    B. 60°
    C. 120°
    D. 180°

30. At a high frequency, as compared to the normal frequency, the value of capacitive reactance in alternating current circuit becomes

    A. zero.
    B. more.
    C. less.
    D. infinite.

31. At a high frequency, the alternating current through a capacitor in an alternating current circuit will be

    A. zero.
    B. large.
    C. small.
    D. undetermined.

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32. In the given truth table, the values of x and y are

A	**B**	$\overline{\mathbf{A}}+\mathbf{B}$
0	0	1
0	1	x
1	0	0
1	1	y

A. $x = 0, y = 0$
B. $x = 0, y = 1$
C. $x = 1, y = 0$
D. $x = 1, y = 1$

33. Which of the following uses a reverse biased diode?

A. Rectifier
B. Photocell
C. Photodiode
D. Light emitting diode

34. The device through which an electrical current passes in one direction only is a

A. diode.
B. buffer.
C. resistor.
D. resistor-transistor.

35. The charge carriers inside an n-type semiconductor material are

A. holes.
B. protons.
C. positrons.
D. electrons.

36. If the reverse current of a reverse biased junction increases to its maximum value, then the

A. voltage decreases.
B. voltage becomes zero.
C. diode resistance decreases.
D. diode's efficiency increases.

37. When a p-n junction is reverse biased, then the depletion region of the junction is

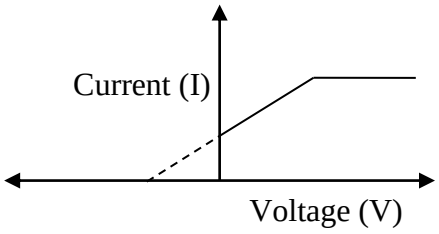
A. zero.
B. widened.
C. narrowed.
D. unchanged.

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38. For non-inverting operational amplifiers, if R_1 is infinite and R_2 is 0 ohm, then the gain of amplifier is

A. –1
B. 0
C. +1
D. infinite

39. The given graph depicts the phenomenon of



A. Compton effect.
B. pair production.
C. photoelectric effect.
D. annihilation of matter.

40. According to Einstein's theory of relativity, gravity is not an ordinary force, but a property of

A. length contraction.
B. space-time interval.
C. space-time geometry.
D. mass-energy relation.

41. Davison and Germer's experiment provided convincing evidences for the

A. wave nature of light.
B. particle nature of light.
C. wave nature of electrons.
D. particle nature of electrons.

42. The theory of non-accelerated frame of reference is called

A. String theory.
B. Special theory of relativity.
C. General theory of relativity.
D. Galilean transformation theory.

43. In all of the following, photocells are used EXCEPT

A. for security systems.
B. for counting systems.
C. in sound track of movies.
D. in air-traffic control stations.

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44. The device that converts light energy into electrical energy is a

A. turbine.
B. dry cell.
C. photocell.
D. generator.

45. Which of the following phenomena demonstrates the wave nature of light?

A. Diffraction
B. Pair production
C. Compton's effect
D. Photoelectric effect

46. The ejection of photoelectrons increases due to an increase in the

A. velocity of photons.
B. threshold frequency.
C. intensity of light source.
D. frequency of light source.

47. When the temperature of a black body is raised, it emits radiations.

With the increase in temperature, the wavelength associated with the maximum intensity

A. becomes zero.
B. remains the same.
C. shifts towards the longer wavelength.
D. shifts towards the shorter wavelength.

48. The phenomena that experimentally proves Einstein's mass-energy relation are

I. Compton's effect
II. photoelectric effect
III. annihilation of matter
IV. pair production of matter

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

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49. According to Heisenberg's uncertainty principle, which of the following option is CORRECT for the position and momentum of a particle?

	Position	Momentum
A	Can be determined precisely	Can be determined precisely
B	Cannot be determined precisely	Can be determined precisely
C	Can be determined precisely	Cannot be determined precisely
D	Cannot be determined precisely	Cannot be determined precisely

50. According to the Special Theory of Relativity, the ratio of apparent mass (m) of an object to its rest mass (m_o) is

A. $\left(1-\frac{v^2}{c^2}\right)^{-2}$

B. $\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}}$

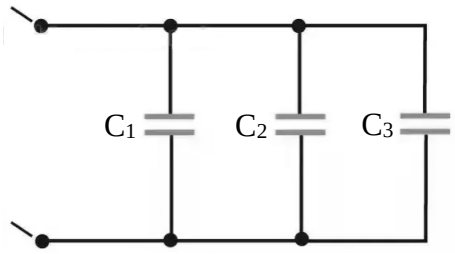
C. $\left(1-\frac{v^2}{c^2}\right)^{\frac{1}{2}}$

D. $\left(1-\frac{v^2}{c^2}\right)^{2}$

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ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

51. Which of the following relations is used to find the equivalent capacitance for the given combination of capacitors?



A. $C_1 + C_2 + C_3$
B. $C_2 / (C_1 + C_3)$
C. $C_1 / (C_2 + C_3)$
D. $1/C_1 + 1/C_2 + 1/C_3$

52. When two or more capacitors are connected in series, then their equivalent capacitance as compared to their individual capacitance will

A. increase.
B. decrease.
C. become zero.
D. remain the same.

53. In an electric circuit, the initial electric current at the beginning of charging a capacitor

A. is zero.
B. is low.
C. is high.
D. cannot be determined.

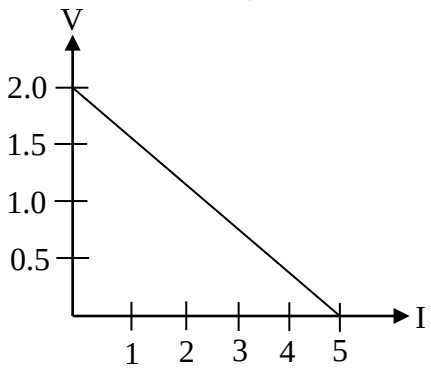
54. With the help of a slide wire bridge, the greatest accuracy will be achieved in measuring an unknown resistance, when

A. the most sensitive galvanometer is used.
B. a varying voltage is provided by a source.
C. a known high resistance box is used in the circuit.
D. the balance point is close to the middle of the wire.

55. For the measurement of potential difference in an electric circuit, a potentiometer is preferred over a voltmeter because the

A. potentiometer has a very low cost.
B. potentiometer is less sensitive than a voltmeter.
C. potentiometer does not take current from the circuit.
D. internal resistance of the potentiometer is more than a voltmeter.

56. The given graph shows the relationship between the potential difference (V) across the terminals of the cell and the current (I) drawn from the cell.



The electromotive force (E) and the internal resistance (r) of the cell are

A. E = 2 V and r = 0.4 Ω.
B. E = 0.4 V and r = 2 Ω.
C. E = 0.5 V and r = 0.4 Ω.
D. E = 0 V and r = 2.5 Ω.

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57. In a physics laboratory, the length of the wire of a potentiometer is 1000 mm and it can measure upto 2 V.

If a standard dry cell of electromotive force 1.5 V obtains a balance at 750 mm, then the electromotive force for a test cell that obtains a balance at 900 mm will be

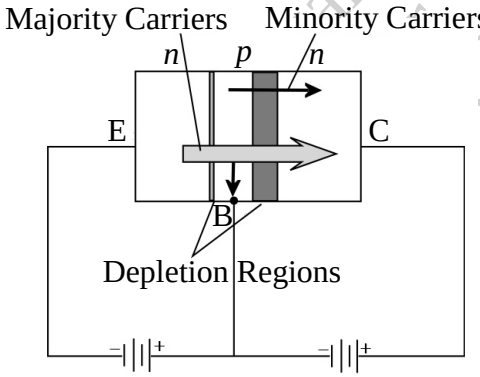
A. 1.80 V.
B. 1.25 V.
C. 0.8 V.
D. 0.5 V.

58. A student is handling a potentiometer that has 10 wires and the balance point is obtained on the 7th wire.

To shift the balance point of the potentiometer to the 9th wire, the resistance of the potentiometer should be

A. increased.
B. decreased.
C. fixed at zero.
D. kept constant.

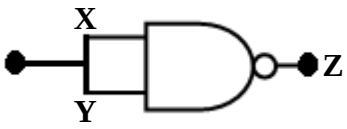
59. In the given NPN transistor diagram, the minority carriers are



A. ions.
B. holes.
C. electrons.
D. molecules.

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60. A student gives input to a NAND gate as shown in the given diagram.



The option that shows the CORRECT truth table for the NAND gate with this arrangement of input is

A

X	Y	Z
0	0	1
0	1	1

B

X	Y	Z
0	0	0
0	1	0

C

X	Y	Z
0	0	1
1	1	0

D

X	Y	Z
0	0	0
1	1	1

61. Which of the following truth tables represents the behaviour of a NAND gate in a burglar alarm?

A

X	Y	Output
0	0	1
0	1	0
1	0	0
1	1	0

B

X	Y	Output
0	0	0
0	1	1
1	0	1
1	1	0

C

X	Y	Output
0	0	0
0	1	0
1	0	0
1	1	1

D

X	Y	Output
0	0	1
0	1	1
1	0	1
1	1	0

62. As the temperature of a transistor rises, the base-emitter resistance will

A. increase.
B. decrease.
C. become zero.
D. remain the same.

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63. When an ideal diode is forward biased in an electric circuit, it behaves as a/ an

A. insulator.
B. conductor.
C. semiconductor.
D. super conductor.

64. When a light source is kept at a distance of 50 cm from a photocell, the stopping potential value is recorded as 0.5 V.

If the distance between the light source and the photocell increases upto 80 cm, then the value of stopping potential will be

A. 0.25 V.
B. 0.5 V.
C. 1.0 V.
D. 2.0 V.

65. In photoelectric effect, the quantity that does NOT affect the work function is the

A. intensity of incident light.
B. nature of emitter material.
C. frequency of incident light.
D. potential difference in the power source.

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